TEXORA · LECTURE SHEET

Yarn Tensile Strength

Yarn strength is where every earlier property — fibre strength, length, fineness, maturity, twist — is finally cashed out. This sheet covers single-yarn and lea (skein) testing, tenacity and the count–strength product, and how to read a strength result like a spinner.

TE-3103 · QUALITY EVALUATION OF FIBRE AND YARN · TOPIC 12 · LEVEL 3 TERM 1

1. The two scales of yarn strength testing

	SINGLE-YARN TEST	LEA (SKEIN) TEST
Specimen	One strand, usually 50 cm gauge	A lea of 120 yd — 80 wraps of yarn acting in parallel (legacy: the lea is broken as one bundle)
Results	Breaking load, elongation %, tenacity, CV% — full distribution incl. weak places	Lea breaking load (lb or kg) only
Character	Modern, informative; the weak places that stop looms appear in its distribution	Traditional, fast; averaging over 80 strands hides weak places — one weak strand sheds its load to its neighbours
Main use	Specification, research, weaving performance prediction	Routine mill control and the classical CSP (below)

Both are performed on CRE (constant rate of extension) testers in modern laboratories — pendulum (inclined-plane) lea testers survive in mill labs — on conditioned yarn (65 % RH, 20 °C; Topic 1), with traverse rate set so breaks occur in the standard time window (commonly 20 ± 3 s for single-yarn CRE tests).

2. Single-yarn strength — quantities

$$\text{Tenacity (cN/tex)} = \text{breaking load (cN)} / \text{yarn linear density (tex)}$$

$$\text{Elongation at break \%} = (\text{extension at break} / \text{original gauge length}) \times 100$$

WORKED EXAMPLE — SINGLE-YARN TENACITY AND ELONGATION

A yarn of 19.7 tex (Ne 30) breaks at 315 cN with an extension of 28 mm on a 500 mm gauge. Tenacity = 315 / 19.7 = **16.0 cN/tex** — a respectable ring-spun cotton figure (typical range roughly 13–18 cN/tex for carded/combed qualities).
Elongation at break = 28/500 × 100 = **5.6 %** — normal for cotton yarn (≈ 5–7 %).
Report with it the CV% of breaking load: at, say, 9 % this yarn is uniform; the same mean with CV 16 % hides weak places that will fail on the loom although "the strength" is unchanged.

3. Lea strength and the count–strength product (CSP)

Because lea strength alone rises with coarseness, the trade combined it with count into a single quality index:

$$\text{CSP} = \text{lea strength (lb)} \times \text{yarn count (Ne)}$$

WORKED EXAMPLE — CSP AND ITS READING

A lea of Ne 20 yarn breaks at 78 lb. CSP = 78 × 20 = **1560**.
Guide values for cotton yarns (mill convention): below ~1800 weak-to-poor for warp service, 1800–2200 serviceable, 2200–2600 good, above ~2600 excellent combed quality — the bands shift with count range and end use, so a mill fixes its own standards per product. CSP 1560 at Ne 20 warns that this yarn, whatever its count accuracy, will give ends-down trouble as warp; the investigation goes back through fibre strength (Topic 8), maturity (Topic 5) and twist (Topic 11) in that order.

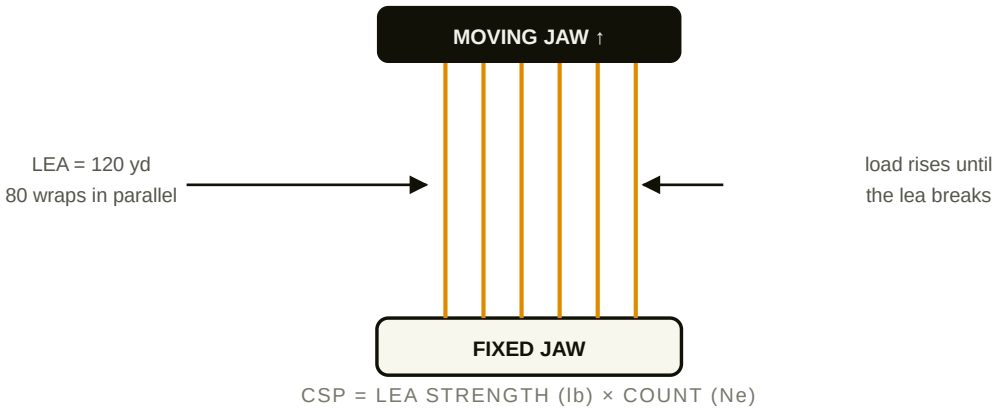


Figure 1 — Lea strength tester principle. The two jaws separate at a constant rate; the whole 120 yd lea is loaded as one parallel bundle.

4. Factors that decide yarn strength

- **Fibre tenacity** (Topic 8) — the ceiling; yarn tenacity typically reaches only about half of fibre tenacity in cotton.
- **Fibre length and short fibre content** (Topic 3) — short fibres slip instead of breaking and contribute little; they also create the thin places where breaks start.
- **Fibre fineness** (Topic 5) — finer fibres put more fibres in the cross-section, sharing load and evening out weak spots.
- **Twist** (Topic 11) — up to its optimum; both under- and over-twisted yarns are weaker.
- **Evenness** (Topic 13) — a yarn breaks at its weakest tested length; mean strength is a fiction if the distribution has a long weak tail. The 5th-percentile strength predicts weaving breaks better than the mean.
- **Count itself** — finer yarns are weaker in load (fewer fibres) at equal tenacity; compare yarns by tenacity or CSP, never by raw breaking load.

5. Testing practice and reporting

- Test at least 30–50 single-yarn breaks (or 10+ leas) per lot, drawn from several packages (Topic 1 sampling).
- Wind off and discard the outer layers of each package before taking specimens — surface yarn is abraded and dried by handling.
- Reject jaw breaks (break within ~5 mm of a jaw in single-yarn tests) and re-test.
- Report: mean breaking load, tenacity, elongation %, CV%, number of tests, gauge length, traverse rate, atmosphere. A bare "strength = 315 cN" answers nothing.
- Keep an eye on the **strength–count relation over time**: a mill's control chart of CSP by count is the quickest health check of the whole spinning line.

Glossary

Lea strength — breaking load of a 120 yd skein tested as one bundle. **CSP** — count–strength product: lea strength (lb) × Ne. **Tenacity** — breaking load per tex. **Elongation at break** — extension at rupture as % of gauge length. **CRE tester** — constant-rate-of-extension strength tester. **Jaw break** — an invalid break at the clamp. **Weak place** — a local section of abnormally low strength, the usual origin of processing breaks.

Key points

• Single-yarn tests reveal distributions and weak places; lea tests give a fast average for routine control.
• Compare yarns by tenacity (cN/tex) or by CSP — never by breaking load across counts.

• CSP = lea lb × Ne; fix interpretation bands per product from your own mill data.
• Strength is the integral of the course: fibre tenacity × length × fineness × twist × evenness.
• Report the CV% and test conditions with every strength mean.

Practice

1. A 25 tex yarn breaks at 400 cN. Find its tenacity and classify it against the typical cotton range given in this sheet.
2. Lea strengths: Ne 24 yarn breaks at 82 lb; Ne 30 yarn at 68 lb. Compute both CSPs and state which is the better yarn.
3. Why does a lea test hide weak places that a single-yarn test exposes? Use the load-sharing idea.
4. A yarn's mean single-yarn strength is satisfactory but its CV% has doubled in a month. Predict the weaving outcome and name two process causes to inspect.

Answer key

1. Tenacity = 400/25 = **16.0 cN/tex** — within, indeed at the good end of, the typical 13–18 cN/tex cotton range.
2. CSP(24) = 82 × 24 = **1968**. CSP(30) = 68 × 30 = **2040**. The Ne 30 yarn is the better yarn although its lea breaks at the lower load — it carries more strength per unit of fineness.
3. In a lea, 80 strands share the load in parallel. When the weakest strand breaks, its share is redistributed among 79 survivors; the recorded lea load is close to a bundle average. A loom, however, loads warp ends singly — each end meets its own weakest place alone. Single-yarn testing reproduces that geometry, so its weak tail is the relevant statistic.
4. Weaving ends-down will rise sharply while routine (lea/mean) reports still look acceptable. Inspect for causes that widen variation: a faulty drafting element on some frames (worn aprons/cots, damaged rollers) and a variable raw-material feed (a new mixing component with high short-fibre content entering unevenly).